



Silver-layer table schemas

Schemas for the three silver-layer source tables referenced in the case study. No data is provided — schema reference only. The view SQL (provided separately) shows how `reporting.view_funnel_dashboard` joins these together.

`silver.pes_subscriptions`

The subscription table from PES (Office Timeline's billing system). One row per subscription event — subscriptions are versioned over time (creation, renewals, seat changes, etc.), so a single subscription has multiple rows. The `ingestion_timestamp` records when the row was ingested.

Column	Type	Description
<code>id</code>	STRING (UUID)	Subscription ID. Stable across versions of the same subscription.
<code>subscription_tracker</code>	STRING (UUID)	Stable identifier that links a subscription chain across co-term events. Post-ABL (Nov 2025) subscriptions only; NULL for older subscriptions.
<code>reference</code>	STRING	Product key. The legacy identifier used for pre-ABL subscriptions. NULL for ABL-migrated subscriptions.
<code>administrator_id</code>	STRING (UUID)	PES user ID of the subscription administrator. Application-layer ID; not directly joinable to HubSpot.
<code>customer_email</code>	STRING	Email of the subscription administrator at the time of the event. May change across rows for the same subscription if the admin is reassigned.
<code>quantity</code>	INT	Seat count for this subscription event. Interpretation depends on <code>orderline_type</code> .

Column	Type	Description
<code>start_date</code>	DATE	Effective start date of this subscription event.
<code>expiry_date</code>	DATE	Effective expiry date. Used to derive active/inactive status (PES has no native active/cancelled flag).
<code>orderline_type</code>	STRING	One of: <code>Create</code> , <code>Extend</code> , <code>AddLicenses</code> , <code>Upgrade</code> , <code>NewCoTermSubscription</code> , <code>CoTermedSubscription</code> , <code>NewSubscriptionAfterCoTerm</code> . Determines how <code>quantity</code> and pricing fields should be interpreted.
<code>discriminator</code>	STRING	<code>CustomerBillingInformation</code> , <code>ResellerBillingInformation</code> , or NULL.
<code>ingestion_timestamp</code>	TIMESTAMP	When this row was ingested. Use <code>ROW_NUMBER() OVER (PARTITION BY id ORDER BY ingestion_timestamp DESC)</code> filtered to <code>rn = 1</code> to get latest state per subscription.
<code>resellers_customer_id</code>	STRING (UUID)	PES user ID of the reseller's named contact, when applicable. NULL for direct purchases. Not stable across coterm chains.
<code>created_at</code>	TIMESTAMP	When this subscription event occurred.

Note on identifiers: `administrator_id` is a PES application-layer user ID. There is no native column on this table that joins to HubSpot's contact identifier. The `customer_email` field is the most direct bridge to HubSpot, but admin emails change over time.

silver.pes_billingdata

Billing records — one row per billing event (purchase, renewal, refund, etc.). Each row is associated with a subscription via `subscription_id` and an order via `order_id`.

Column	Type	Description
<code>order_id</code>	STRING (UUID)	Order identifier. One order may contain multiple billing rows (e.g., a multi-line invoice).
<code>subscription_id</code>	STRING (UUID)	Joins to <code>pes_subscriptions.id</code> .

Column	Type	Description
<code>customer_email</code>	STRING	Customer email for the billing event.
<code>email</code>	STRING	Buyer email. May differ from <code>customer_email</code> (e.g., when procurement places the order on behalf of an admin).
<code>license_manager_email</code>	STRING	Email of the license manager at the time of billing. May differ from both <code>customer_email</code> and <code>email</code> .
<code>discriminator</code>	STRING	<code>CustomerBillingInformation</code> , <code>ResellerBillingInformation</code> , or NULL.
<code>created_at</code>	TIMESTAMP	When the billing event occurred.
<code>gross_amount</code>	DECIMAL	Gross billing amount before discounts and taxes.
<code>currency</code>	STRING	ISO currency code.

Note on identifiers: Three different email fields can point to the same underlying customer/company. None of them join cleanly to HubSpot's `email` field without normalization.

silver.hs_otl_contacts

HubSpot contacts (raw silver layer). One row per HubSpot contact ID. CDC'd from HubSpot, so contact properties may differ from the latest values in HubSpot if there's ingestion lag.

Column	Type	Description
<code>id</code>	STRING	HubSpot contact ID. Primary key.
<code>email</code>	STRING	Contact email at time of ingestion. May contain mixed case.
<code>created_date</code>	TIMESTAMP	When the HubSpot contact was created.
<code>lifecycle_stage</code>	STRING	HubSpot lifecycle stage (subscriber, lead, customer, etc.).
<code>original_traffic_source</code>	STRING	HubSpot first-touch attribution: <code>Paid Search</code> , <code>Organic Search</code> , <code>Direct Traffic</code> , <code>AI Referrals</code> , <code>Referrals</code> , <code>Email</code>

Column	Type	Description
		Marketing , Social Media , Other Campaigns , Paid Social , Offline .
contact_source_form	STRING	Form that created the contact. See data dictionary for value reference.
contact_source_type	STRING	Channel category. Form , Integration , Import , Conversations , Crm_ui , Api , Bcc_to_crm , Meetings .
hs_lead_status	STRING	HubSpot lead status (NEW, OPEN, QUALIFIED, etc.).
company_id	STRING	HubSpot company ID, if associated. May be NULL for contacts without a company association.
country	STRING	Country derived from contact properties.
is_buyer	INT	Label-based heuristic. Set to 1 if the contact's HubSpot labels (joined from hs_otl_labels) contain 'resellers customer' or 'owner' . Independent of whether the contact has any associated paid record.
pes_user_guid	STRING (UUID)	PES user GUID, if a join was successfully made. NULL for the majority of HubSpot contacts , including most contacts with completed installs and purchases — the silver-to-gold transform fails to resolve this field for most users (see the Gold Layer NULL-rate stats page).
ingestion_timestamp	TIMESTAMP	When this row was ingested.

Note on identifiers: pes_user_guid is the column intended to bridge HubSpot to PES. Its NULL rate is the central problem motivating Part 3.

How the view joins them

The full SQL of reporting.view_funnel_dashboard is provided separately. At a high level:

1. Starts from hs_otl_contacts

2. Bridges contacts to PES activity via `tracking_id` (sourced from `pes_download_histories` keyed on email) for install events, license creation, and subscription creation paths
3. Bridges contacts to PES purchases via direct email match (`hs_otl_contacts.email` → `view_pes_transactions.email`)
4. Computes the funnel-stage flags (`has_download` , `has_start_install` , `has_completed_install` , `has_purchased` , etc.) at the contact level
5. Applies `ROW_NUMBER()` to keep one row per contact

Notably, the view does its own ad-hoc identity resolution rather than relying on a canonical user dimension in gold. That's the structural problem Part 3 asks you to address.